MIL-C-20367G
23 January 1986
SUPERSEDING
MIL-C-20367F
20 February 1976

MILITARY SPECIFICATION

CASE, CARTRIDGE, 20MM: M21A1

This specification is approved for use by the all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the M21A1 Cartridge Case for use in the assembly of 20mm cartridges.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specifications For

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-651	-	Visual Inspection Standards for 20MM Ammunition and Components
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1305
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2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.7)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

PRODUCT AND PACKAGING DRAWINGS

7258849	-	Cartridge, 20mm, Target Practice, M99A1
7258856	-	Carton, Interplant Shipment for 20mm, Cartridge Cases, Empty, Unprimed
7259106	-	Cartridge, 20mm, Target Practice, M204
7553197	-	Case, Cartridge, 20mm, M21A1
IL7553197	-	Index of Inspection Equipment Lists for: Case, Cartridge, 20mm, M21A1
7553663	-	Classification of Firing Casualties for: Case, Cartridge, 20mm, M21A1 and M21A1B1

INSPECTION EQUIPMENT DRAWINGS

5055700	-	Gage, Snap
5055701	-	Gage, Progressive Snap
5055702	-	Gage, Length
5055709	-	Gage, Snap
6023715	-	Gage, Snap
7259434	-	Scale
7288637	-	Gage, Indicator Depth
7295157	-	Gage, Dial Indicator
7297557	-	Gage, Indicator
7299969	-	Gage, Indicator
7299996	-	Gage, Indicator
7558527	-	Gage, Chamber

PUBLICATIONS

US ARMY MATERIAL COMMAND

AMCR 715-505	-	Ammunition Ballistic Acceptance Test methods, Volume 8, Test Procedures for 20mm Cartridges

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.1.3 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E18 - Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General.

a. Materials. Materials shall be in accordance with the applicable drawings and specifications.

b. Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without prior approval (6.4).

3.2 Cartridge case. The cartridge case shall comply with the requirements specified on Dwg. 7553197 as applicable, referenced specifications and other requirements specified herein.

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Hardness. The cartridge case shall be in accordance with the hardness requirements of Dwg. 7553197.

3.5 Residual stresses. The cartridge case shall be free of residual stresses to the degree detectible by the mercurous nitrate test.

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3.6 Cleaning. Sodium dichromate, or similar solutions commonly used to obtain a bright finish on brass products, is prohibited from use in the cleaning process for the cartridge cases.

3.7 Neutrality. The cartridge case surfaces shall be free of acids and alkalis.

3.8 Function and casualty. The cartridge case shall function without casualty.

3.9 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.9.1 Metal processing and handling defects. The cartridge case shall be free of defects such as metal flaws, folds, wrinkles, splits, cracks, dents, and perforations.

3.9.2 Cleanliness. The cartridge case shall be free of foreign matter such as corrosion, grease, oil, corrosive residue, stains, dirt, lint, chips, slivers, shavings and trim sections.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection (4.3)

b. Quality Conformance Inspection (4.4)

4.3 First article inspection.

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4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Parts description	Drawings	Quantity
a. Inspection and Non Firing Tests: Case, Cartridge	7553197	125
b. Firing Tests: Case, Cartridge	7553197	4000

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078. Failure of the original first article sample to be in accordance with the applicable drawings and requirements of Table I, as specified herein shall result in sample disapproval.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-C-20367G

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
	Case, Cartridge, 20mm, M21A1		1 1		7553197
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Case, Cartridge, 20mm, M21A1				
	A. Dimensional and non-firing Tests				
	Examination of defects	125	Acc 0 Rej 1	3.2	4.4.2.1 1/
	Hardness	5	Acc 0 Rej 1	3.4	4.5.1 1/
	Residual stress	25	Acc 0 Rej 1	3.5	4.5.2 1/
	Neutrality	10	Acc 0 Rej 1	3.7	4.5.3 1/
	B. Function and Casualty				
	Examination of defects	4000	Acc 0 Rej 1	3.2/3.8	4.4.2.1/4.5.4

NOTES:

1/ Tests shall be conducted using same sample cases.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. To determine product acceptability, major and minor defects as listed herein may be considered collectively on a class basis or defects may be considered individually. When a class basis is utilized the Acceptable Quality Level (AQL) for major defects shall be 1.50 percent and the AQL for minor defects shall be 2.50 percent. However, when major defects are considered collectively on a class basis, the AQL for any individual defect shall be limited to .40 percent. In addition, when three or less defects appear in a classification sampling plan applicable to a class shall not be used for major defects. (See 6.6)

c. MIL-STD-651 and Dwg. 7553663 shall apply in defining and evaluating cartridge visual defects.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER
4.4.2.1	Case, Cartridge, 20mm, M21A1		7553197
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101.	Flash hole missing or obstructed		100%	3.2	Visual
102.	Thickness, head rim		.25%	3.2	6023715
103.	Chamfer, head rim, missing		.25%	3.2	Visual
104.	Diameter, head, min.		.25%	3.2	5055700
105.	Diameter, primer pocket		.25%	3.2	Gage
106.	Depth, primer pocket		.25%	3.2	7288637
107.	Runout, primer pocket with datum surface		.25%	3.2	Gage
108.	Diameter, extractor groove, max		.25%	3.2	5055701
109.	Width, extractor groove, min		.25%	3.2	5055702
110.	Wall thickness, at first location from inside head surface (set block .045)		.25%	3.2	7299996
111.	Wall thickness, at second location from inside head surface (set block .030)		.25%	3.2	7299996
112.	Inside diameter, mouth, min		.25%	3.2	Gage
113.	Profile and alignment, max		.25%	3.2	7558527

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-20367G

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 7553197
4.4.2.1	Case, Cartridge, 20mm, M21A1				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
114.	Length to .876 basic diameter, min		.25%	3.2	7297557
115.	Total length, max		.25%	3.2	5055709
116.	Diameter, body taper at location near head, min		.25%	3.2	Gage
117.	Diameter, body taper at 2.023 location, min		.25%	3.2	Gage
118.	Diameter, body taper at location near shoulder, min		.25%	3.2	Gage
119.	Head configuration		.25%	3.2	Visual
120.	Metal defective		.25%	3.9.1	Visual
121.	Corrosion		.25%	3.9.2	Visual
122.	Oil or Grease		.25%	3.9.2	Visual
MINOR:					
201.	Wall thickness at mouth (set block .028)		.40%	3.2	7259157
202.	Total length, min		.40%	3.2	5055709
203.	Diameter, extractor groove, min		.40%	3.2	5055701
204.	Width, extractor groove, max		.40%	3.2	5055702
205.	Base thickness (set block .387)		.40%	3.2	7299969
206.	Diameter, flash hole		.40%	3.2	Gage
207.	Diameter, head, max		.40%	3.2	5055700

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Case, Cartridge, 20mm, M21A1	3 OF 3	7553197
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
208.	Inside diameter, mouth, max		.40%	3.2	Gage
209.	Weight		.40%	3.2	Balance/7259434
210.	Length to .876 basic diameter, max		.40%	3.2	7297557
211.	Radius at mouth of primer pocket missing or not tangent to pocket sidewall		.40%	3.2	Visual
212.	Marking incorrect, incomplete or unidentifiable		.40%	3.2	Visual
213.	Scratch (other than draw)		.40%	3.9.1	Visual
214.	Fold or wrinkle		.40%	3.9.1	Visual
215.	Dent		.40%	3.9.1	Visual
216.	Foreign matter (other than oil, grease or corrosion)		.40%	3.9.2	Visual
217.	Evidence of poor workmanship		.65%	3.9	Visual

NOTES:

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4.4.3 Tests. The tests listed in Table II shall be performed on each cartridge case lot in accordance with the test methods prescribed in 4.5. Sample size and acceptance criteria for each test shall be as specified. Only cartridge cases having met the visual and dimensional requirements and having been selected in such a manner that the sample is representative of the entire lot shall be used in the tests. The selected cartridge cases shall be thoroughly mixed before being divided into samples for the various tests.

TABLE II. CARTRIDGE CASE TESTS

Test		Sample	Requirement/Test Paragraph
Hardness	1/	5	3.4/4.5.1
Residual Stress	2/	25	3.5/4.5.2
Neutrality	3/	10	3.7/4.5.3
Function and casualty	4/	200	3.8/4.5.4

1/ Failure of two or more of the sample cartridge cases to conform to the hardness requirements shall cause rejection of the lot. If one of the samples fails to conform, a second sample of 10 cartridge cases shall be randomly selected from the lot and tested. The lot shall be rejected if a total of two or more cartridge cases of the combined first and second samples fails to conform to the applicable hardness requirement. Failure is defined as any individual hardness number obtained that does not conform to the drawing hardness pattern position requirements.

2/ The lot shall be rejected if one or more cartridge cases of the sample split.

3/ The lot shall be rejected if one or more cartridge cases of the sample show evidence of acidity or alkalinity.

4/ The lot shall be rejected if any malfunction or casualty of Table III occurs in number(s) equal to or greater than the applicable "Rej" number. If malfunctions or casualties occur in excess of the applicable "Acc" number, but less than the applicable "Rej" number, a second sample of 400 cartridge cases randomly selected from the lot, shall be tested. The lot shall be rejected if in the combined samples malfunctions or casualties occur in number(s) equal to or greater than the applicable "Rej" number.

4.4.3.1 Firing defects. Firing defects and associated criteria shall be as specified in Table III. For defect definitions, see AMCR 715-505, Volume 8.

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TABLE III. FIRING DEFECTS

Firing Casualties		Acc	Rej
Failure to chamber	1/	0	1
Failure to extract	2/	0	1
Blown primer	3/	0	1
Splits and ruptures	4/		
Splits:			
I or S		6	17
J		2	7
K, L, or M		0	1
Rupture, partial			
S, J, or K		1	6
L		0	1
Circumferential rupture, complete		0	1
Detached metal	5/	0	1

1/ The sample cartridge case shall be classed defective only if it can be shown that failure of the bolt to close fully on the cartridge was caused by a cartridge case defect.

2/ A fired sample cartridge case remaining in the chamber after the normal extraction cycle shall be classed defective, unless it can be shown that the malfunction was caused by a defective weapon.

3/ The sample cartridge case shall be classed defective only if it is evident by visual inspection that both the primer pocket and case head are enlarged and deformed.

4/ Refer to Dwg. 7553663 and AMCR 715-505, Volume 8, for classifying splits and ruptures resulting from the function and casualty test. If a split or partial rupture extends into two or more defined areas, only the most severe defect criterion of Table III for the areas involved shall be used.

5/ The case lot shall not be penalized if this malfunction cannot be shown to have resulted from a defective case. There must be evidence of detached metal from the fired case exterior. The lot will not be penalized for the detachment of metal shavings from the projectile crimped area of the case interior.

4.4.3.2 Unlisted firing defects. The lot shall be suspended and referred to the contracting officer for disposition, if a malfunction or casualty not covered by this specification occurs in any firing test.

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4.4.4 Packaging, packing, marking and palletization. Inspection for packaging, packing, marking and palletization shall comply with Dwg. 7258856 and with applicable requirements of MIL-STD-1169.

4.4.5 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Index of Inspection Equipment List-LI 7553197 and the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2. The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.3 herein.

4.5 Methods of inspection.

4.5.1 Hardness. The cases of the test sample(s) shall be sectioned as shown on Dwg. 7553197. The head sections shall have approximately parallel surfaces finished to 63 microinches or finer, care being taken not to change the hardness in obtaining this finish. Four determinations each shall be made at the indicated body and neck positions for each sample case, two on each half on the interior as shown. Two determinations each shall be made at the indicated head position as shown for each sample case. The method of test shall be as specified in ASTM E18 - Method of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

4.5.2 Residual stresses. The method of test shall be as specified in AMCR 715-505, Volume 8, for case mercurous nitrate test.

4.5.3 Neutrality. The method of testing cartridge cases for acidity and alkalinity shall be as follows using red and blue sensitive litmus paper. Neutral distilled water shall be used to moisten the paper and the case to provide adhesion. The litmus paper shall be handled with forceps to prevent contamination from fingers. Smoking should not be permitted in the vicinity of test since tobacco smoke will turn red litmus blue. Pieces of red and blue sensitive litmus paper will be placed in several locations both on the interior and exterior of the case. A definite color change is defined as the color produced when blue litmus is immersed for thirty seconds in one thousandth normal (.001N) hydrochloric or sulfuric acid, or when red litmus is immersed for thirty seconds in one thousandth normal (.001N) sodium hydroxide. Litmus paper is considered of sufficient sensitivity if it shows change in color when immersed for 45 seconds, with stirring, in five ten thousandths normal (.0005N) solutions of the corresponding acid or alkali mentioned.

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4.5.4 Function and casualty. The method of test shall be as specified in AMCR 715-505, Volume 8. The test cases shall be assembled into cartridges, either target practice, 20mm, M99A1, Dwg. 7258849, and fired in an M24A1 gun or target practice, 20mm, M204, Dwg. 7259106, and fired in an M3 gun, in bursts of 50 rounds at a cyclic rate of approximately 750 rounds per minute. The gun barrel used shall be at ambient temperature at the beginning of the test and will be cooled to ambient temperature after each 100 rounds. This test shall be conducted at a facility designated by the procuring activity. In order to minimize inspection costs, the function and casualty test shall be performed after the sample has been found acceptable for all other requirements.

5. PACKAGING

5.1 Preservation and packaging. The cartridge cases shall be preserved and packaged in accordance with Dwg. 7258856, as directed by the procuring activity.

5.2 Packing. The cartridge cases shall be preserved and packaged as specified in 5.1. Each lot shall be packed for shipment in accordance with Dwg. 7258856, as directed by the procuring activity.

5.3 Marking. Marking or labeling of packed cartons shall comply with the applicable requirements of MIL-STD-1169.

5.4 Palletization. Palletization of packed cartons shall comply with the applicable requirements of Dwg. 7258856.

6. NOTES

6.1 Intended use. The cartridge cases covered by this specification are intended for use in 20mm automatic gun systems that have been designed for firing cartridges having the M21A1 case configuration.

6.2. Ordering data. See MIL-A-48078.

6.3. Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARDC, ATTN: AMSMC-QAF-I(D), Dover, NJ, 07801-5001.

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6.4 Process deviation. A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which alters the metallurgical or physical properties of the item.

6.5 First article sample. In order to minimize inspection costs, the firing tests will be performed after the sample has been provisionally accepted by the government for all other requirements. Additional cases may be required by the test facility.

6.6 Intermediate point inspection. The classification of defects identifies the defect characteristics for acceptance inspection. It may be necessary to modify the sequence of inspection stations to best suit the manufacturing process. Inspection for defect characteristics which will be hidden or altered by subsequent processing operations (including unrelated operations), should be scheduled to prevent premature acceptance which could be detrimental to the attainment of optimum product quality of the end item.

6.7 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or U.S. Army Armament Research and Development Command (ARRADCOM) drawings. Technical data originally prepared by these activities are now under the cognizance of ARDC.

6.8 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

Review activities:
Army-AR
Air Force-70

(Project 1305-0B20)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-20367G	CASE, CARTRIDGE, 20MM: M21A1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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